

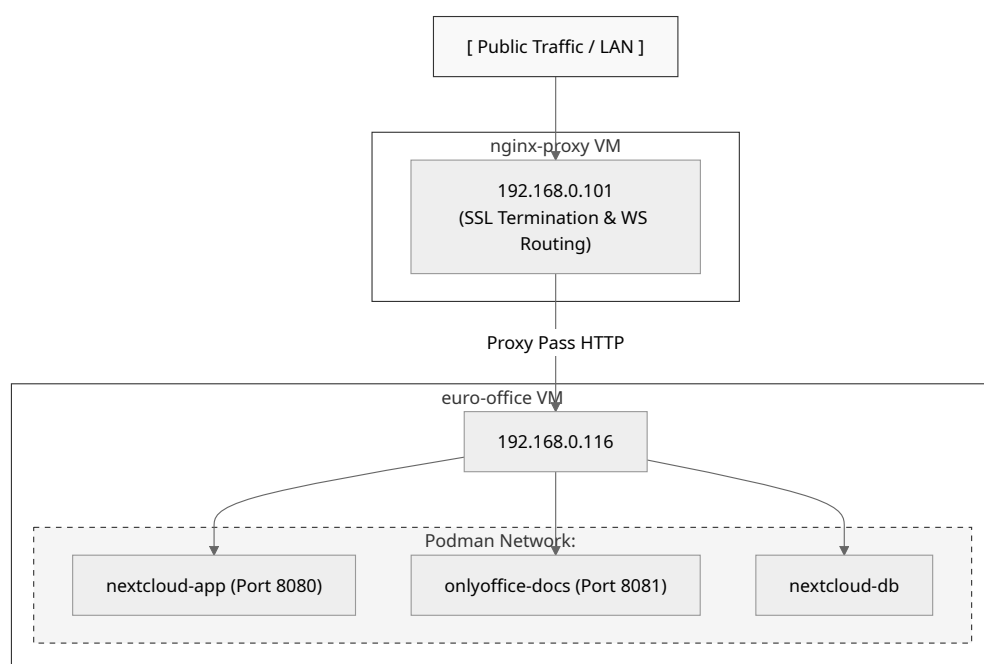
Sovereign Digital Workspace Setup Guide

Nextcloud & ONLYOFFICE Docs with Rootless Podman and Nginx Reverse Proxy

This document provides a step-by-step blueprint to deploy a sovereign collaborative office suite using rootless Podman and Fedora-native Quadlets on a single VM, proxied securely behind an independent Nginx instance.

Prerequisites & Architecture

Error rendering diagram: Parse error on line 19: ...://onlyoffice-docs/ (Internal Mesh)|
OO_ -----^ Expecting 'SQE', 'DOUBLECIRCLEEND', 'PE', '-)', 'STADIUMEND',
'SUBROUTINEEND', 'PIPE', 'CYLINDEREND', 'DIAMOND_STOP', 'TAGEND', 'TRAPEND',
'INVTRAPEND', 'UNICODE_TEXT', 'TEXT', 'TAGSTART', got 'PS'



- Application VM (Host OS): Fedora Server
- Hostname: euro-office
 - IP Address: 192.168.0.116
- Reverse Proxy VM: Nginx Server
- Hostname: nginx-proxy
 - IP Address: 192.168.0.101

- Internal Port Allocations (on euro-office):
 - Nextcloud Core Web Access: Port 8080
 - ONLYOFFICE Document Server Access: Port 8081
-

Part 1: Nginx Proxy Configuration (nginx-proxy VM)

Log into the nginx-proxy (192.168.0.101) server.

Create a virtual host configuration file to route traffic to the application VM.

Replace **://yourdomain.com** with the actual domain names.

On Ubuntu/Debian systems, the configuration file is located at **/etc/nginx/sites-available/euro-office.conf** and enabled via a symbolic link to **/etc/nginx/sites-enabled/**.

On RHEL/Fedora/Rocky Linux systems, the file should be created directly at **/etc/nginx/conf.d/euro-office.conf**.

1. Nextcloud Frontend Proxy Block

```
# 1. Define your domain name environment variable
export MY_DOMAIN="gardenofrot.cc"

# 2. Overwrite office.conf with modern HTTP/2 directives
sudo tee /etc/nginx/sites-available/office.conf > /dev/null << EOF
# =====
# 1. NEXTCLOUD FRONTEND PROXY BLOCK (Port 8080)
# =====
server {
    listen 80;
    listen [::]:80;
    server_name nextcloud.${MY_DOMAIN};
    return 301 https://\${host}\${request_uri};
}

server {
    listen 443 ssl;
    listen [::]:443 ssl;
    server_name nextcloud.${MY_DOMAIN};

    # Modern HTTP/2 Activation Directive
    http2 on;

    # SSL Certificate Paths
    ssl_certificate /etc/letsencrypt/live/${MY_DOMAIN}/fullchain.pem;
    ssl_certificate_key /etc/letsencrypt/live/${MY_DOMAIN}/privkey.pem;

    ssl_protocols TLSv1.2 TLSv1.3;
    ssl_prefer_server_ciphers off;

    # Performance and Security Headers for large file syncs
    client_max_body_size 10G;
    fastcgi_buffers 64 4K;

    location / {
        proxy_pass http://192.168.0.116:8080;
        proxy_set_header Host \${host};
        proxy_set_header X-Real-IP \${remote_addr};
        proxy_set_header X-Forwarded-For \${proxy_add_x_forwarded_for};
        proxy_set_header X-Forwarded-Proto \${scheme};

        # Service Discovery Redirects for Nextcloud Sync Clients
        location ^~ /.well-known/carddav { return 301 \${scheme}://\${host}/remote.php/d
av/; }
        location ^~ /.well-known/caldav { return 301 \${scheme}://\${host}/remote.php/d
av/; }
    }
}
```

```
# =====
# 2. ONLYOFFICE DOCUMENT SERVER PROXY BLOCK (Port 8081)
# =====
server {
    listen 80;
    listen [::]:80;
    server_name office.${MY_DOMAIN};
    return 301 https://\${host}\${request_uri};
}

server {
    listen 443 ssl;
    listen [::]:443 ssl;
    server_name office.${MY_DOMAIN};

    # Modern HTTP/2 Activation Directive
    http2 on;

    # SSL Certificate Paths
    ssl_certificate /etc/letsencrypt/live/${MY_DOMAIN}/fullchain.pem;
    ssl_certificate_key /etc/letsencrypt/live/${MY_DOMAIN}/privkey.pem;

    ssl_protocols TLSv1.2 TLSv1.3;
    ssl_prefer_server_ciphers off;

    # Payload limits for large documents
    client_max_body_size 100M;

    location / {
        proxy_pass http://192.168.0.116:8081;

        proxy_redirect http:// \${scheme}://;
        proxy_set_header Host \${host};
        proxy_set_header X-Real-IP \${remote_addr};
        proxy_set_header X-Forwarded-For \${proxy_add_x_forwarded_for};
        proxy_set_header X-Forwarded-Host \${host};
        proxy_set_header X-Forwarded-Proto https;

        # WebSocket support for live collaboration editing
        proxy_http_version 1.1;
        proxy_set_header Upgrade \${http_upgrade};
        proxy_set_header Connection "upgrade";

        # Timeouts to stop disconnects during quiet editing phases
        proxy_read_timeout 3600s;
        proxy_send_timeout 3600s;
        proxy_connect_timeout 10s;
        proxy_next_upstream error timeout invalid_header http_502;
    }
}
```

```
EOF
```

```
# Create the Symbic Link
sudo ln -s /etc/nginx/sites-available/office.conf /etc/nginx/sites-enabled/

# Test he link worked
# 1. Test the Nginx configuration for syntax errors
sudo nginx -t

# 2. If the test passes, reload Nginx to apply changes safely
sudo systemctl reload nginx
```

Part 2: System Initialization & Firewall (euro-office VM)

Log into the euro-office (192.168.0.116) server as a non-root user with sudo privileges.

```
# 1. Update the Debian package tree indices
sudo apt update -y

# 2. Upgrade any outdated system packages
sudo apt upgrade -y

# 3. Install Podman natively on Debian 13
sudo apt install -y podman

# 4. Enable user lingering so containers stay running after SSH logout
sudo loginctl enable-linger $USER

# Install the native Debian firewall manager
sudo apt install -y ufw

# Allow your SSH connection first so you don't get locked out!
sudo ufw allow ssh

# Open the Nextcloud and ONLYOFFICE web ports
sudo ufw allow 8080/tcp
sudo ufw allow 8081/tcp

# Enable the firewall service
sudo ufw --force enable

echo "-> Configuring Debian rootless sub-UID namespaces..."
sudo usermod --add-subuids 100000-165535 --add-subgids 100000-165535 $USER || true
podman system migrate

# 1. Lower the secure port allocation threshold to port 80
sudo sysctl -w net.ipv4.ip_unprivileged_port_start=80

# 2. Make the port override permanent across system reboots
echo "net.ipv4.ip_unprivileged_port_start=80" | sudo tee -a /etc/sysctl.conf

# 3. Force Podman to clear its internal state and sync changes
podman system migrate
```

Part 3: Storage Allocation & User Namespace Adjustment

```
cat << 'EOF' > step3_storage.sh
#!/bin/bash
set -e

echo "=== [Step 3] Initializing Podman Volume Infrastructure ==="

# Cache sudo credentials upfront so the script runs unattended
sudo -v

# List of required volumes
VOLUMES=("nc-db-data" "nc-app-data" "nc-app-config" "oo-data" "oo-logs")

# 1. Destructive check and initialization loop
echo "-> Provisioning Podman volumes (Wiping existing data if found)..."
for vol in "${VOLUMES[@]}; do
    if podman volume exists "$vol"; then
        echo "    [!] Volume '$vol' already exists. Purging and resetting..."
        podman volume rm -f "$vol"
    fi
    podman volume create "$vol"
done

# 2. Extract physical host volume mount directory points
echo "-> Extracting volume mountpoints..."
CONFIG_PATH=$(podman volume inspect nc-app-config --format '{{.Mountpoint}}')
DATA_PATH=$(podman volume inspect nc-app-data --format '{{.Mountpoint}}')

# 3. Shift directory execution permissions to UID 33 (www-data) inside rootless space
echo "-> Realignment of user namespaces for Nextcloud internal engines..."
podman unshare chown -R 33:33 "$CONFIG_PATH"
podman unshare chown -R 33:33 "$DATA_PATH"

echo "=== [Step 3] Storage infrastructure deployed cleanly! ==="
EOF

# Make the script executable
chmod +x step3_storage.sh

# Execute the updated step 3 storage deployment script
./step3_storage.sh
```

Part 4: Provision Quadlet Service

Definitions

Create the automated container layouts using systemd Quadlets.

```
cat << 'EOF' > step4_quadlets.sh
#!/bin/bash
set -e

echo "=== [Step 4] Orchestrating Quadlet Deployment & Lifecycle ==="

# 1. Establish absolute systemd environment linkages for Debian 13
export XDG_RUNTIME_DIR="/run/user/${id -u}"
export DBUS_SESSION_BUS_ADDRESS="unix:path=${XDG_RUNTIME_DIR}/bus"

TARGET_DIR="${HOME}/.config/containers/systemd"
FILES=("nextcloud-db.container" "onlyoffice-docs.container" "nextcloud-app.container")

# =====
# 2. GENERATE STATIC CONFIGURATION FILES (Clean, Decoupled Git-Safe Assets)
# =====
echo "-> Generating static Quadlet configuration files locally..."

# 1. PostgreSQL Database definition (Removed network dot dependency)
cat << 'LOCAL_EOF' > nextcloud-db.container
[Unit]
Description=Nextcloud PostgreSQL Database

[Container]
ContainerName=nextcloud-db
Image=docker.io/library/postgres:15-alpine
Network=nextcloud-net
Volume=nc-db-data:/var/lib/postgresql/data:Z
Environment=POSTGRES_DB=nextcloud POSTGRES_USER=nextcloud_user POSTGRES_PASSWORD=YourSecurePasswordHere

[Install]
WantedBy=default.target
LOCAL_EOF

# 2. ONLYOFFICE Document Server definition
cat << 'LOCAL_EOF' > onlyoffice-docs.container
[Unit]
Description=Sovereign Euro-Office Document Engine
After=nextcloud-net-network.service
Requires=nextcloud-net-network.service

[Container]
ContainerName=onlyoffice-docs
# Swaps upstream ONLYOFFICE for the official European-governed Nextcloud AIO Office suite
Image=docker.io/nextcloud/aio-collabora:latest
Network=nextcloud-net
```

```
PublishPort=8081:9980
Environment=allow_not_encrypted=true

[Install]
WantedBy=default.target
LOCAL_EOF

# 3. Nextcloud Core App Engine definition
cat << 'LOCAL_EOF' > nextcloud-app.container
[Unit]
Description=Nextcloud Application Server
After=nextcloud-db.service onlyoffice-docs.service

[Container]
ContainerName=nextcloud-app
Image=docker.io/library/nextcloud:stable
Network=nextcloud-net
PublishPort=8080:80
Volume=nc-app-data:/var/www/html:Z
Volume=nc-app-config:/var/www/html/config:Z

# FIX: Forces internal loopbacks to resolve locally via the proxy, bypassing Hairpin
# NAT limits
AddHost=office.gardenofrot.cc:192.168.0.101
AddHost=nextcloud.gardenofrot.cc:192.168.0.101

Environment=POSTGRES_HOST=nextcloud-db POSTGRES_DB=nextcloud POSTGRES_USER=nextcloud
_user POSTGRES_PASSWORD=YourSecurePasswordHere NEXTCLOUD_ADMIN_USER=admin NEXTCLOUD_
ADMIN_PASSWORD=YourSecureAdminPassword

[Install]
WantedBy=default.target
LOCAL_EOF

# =====
# 3. NATIVE PODMAN NETWORK ASSURANCE & SYSTEMD INJECTION
# =====

# Stop previous failed service iterations to clear execution locks
echo "-> Hard stopping old environment stacks..."
systemctl --user stop nextcloud-app.service onlyoffice-docs.service nextcloud-db.ser
vice 2>/dev/null || true
systemctl --user reset-failed 2>/dev/null || true

# Explicitly create the Podman user network if it doesn't already exist
echo "-> Verifying and creating rootless network mesh (nextcloud-net)..."
if ! podman network exists nextcloud-net; then
    podman network create nextcloud-net
else
```

```
    echo "    Network mesh already active."
fi

# Refresh target configuration directory paths and purge old broken files
echo "-> Syncing layout files to systemd user paths..."
mkdir -p "$TARGET_DIR"
rm -f "$TARGET_DIR/nextcloud-net.network" # Wipe the old network unit file causing c
onflicts

for file in "${FILES[@]"; do
    cp "$file" "$TARGET_DIR/"
done

# Compile definitions into active systemd services
echo "-> Re-compiling systemd user service structures..."
systemctl --user daemon-reload

# Launch the fully compiled application stack sequentially
echo "-> Launching backend engine stacks..."
systemctl --user start nextcloud-db.service onlyoffice-docs.service

echo "-> Mounting nextcloud frontend interface..."
systemctl --user start nextcloud-app.service

echo "=== [Step 4] Quadlet environment deployed and started cleanly! ==="
EOF

# Ensure script is executable
chmod +x step4_quadlets.sh

# Run the updated runner script
./step4_quadlets.sh
```

```
# 1. Correct the public DocumentServerUrl parameter to use the office subdomain
export MY_DOMAIN="gardenofrot.cc"

echo "-> Linking Nextcloud to the correct Euro-Office subdomain..."
podman exec -u www-data nextcloud-app php occ config:app:set eurooffice DocumentServerUrl --value="https://office.${MY_DOMAIN}/"

# 2. Re-verify the advanced internal routing endpoints are intact
podman exec -u www-data nextcloud-app php occ config:app:set eurooffice DocumentServerInternalUrl --value="http://onlyoffice-docs/"
podman exec -u www-data nextcloud-app php occ config:app:set eurooffice StorageUrl --value="http://nextcloud-app/"

# 1. Establish absolute systemd environment linkages for Debian 13
export XDG_RUNTIME_DIR="/run/user/${id -u}"
export DBUS_SESSION_BUS_ADDRESS="unix:path=${XDG_RUNTIME_DIR}/bus"

echo "=== Aligning Core Security Matrix and Proxy Frames ==="

# 2. Force Nextcloud to strictly trust HTTPS proxy headers globally
podman exec -u www-data nextcloud-app php occ config:system:set overwritehost --value="nextcloud.gardenofrot.cc"
podman exec -u www-data nextcloud-app php occ config:system:set overwriteprotocol --value="https"
podman exec -u www-data nextcloud-app php occ config:system:set overwritecli.url --value="https://nextcloud.gardenofrot.cc/"

# 3. Explicitly whitelist the ONLYOFFICE subdomain inside Nextcloud's security headers
podman exec -u www-data nextcloud-app php occ config:app:set eurooffice intrusion_detection_whitelist --value="https://office.gardenofrot.cc" || true

# 4. Cleanly power-cycle the stack to flush memory states
echo "-> Cycling application containers..."
systemctl --user restart nextcloud-app.service onlyoffice-docs.service
#####

# 1. Establish absolute systemd environment linkages for Debian 13
export XDG_RUNTIME_DIR="/run/user/${id -u}"
export DBUS_SESSION_BUS_ADDRESS="unix:path=${XDG_RUNTIME_DIR}/bus"

echo "-> Executing Mimetype database migration repair loops..."
podman exec -u www-data nextcloud-app php occ maintenance:repair --include-expensive

echo "-> Registering a persistent nightly 2:00 AM maintenance window..."
# Setting the value to 2 maps to 02:00 UTC/Local depending on container clock configurations
podman exec -u www-data nextcloud-app php occ config:system:set maintenance_window_s
```

```
tart --value=2 --type=integer
```

```
#####
```

Nginx

Log into your nginx-proxy VM (192.168.0.101) via SSH, and run this block:

```
# Define your domain name environment variable
export MY_DOMAIN="gardenofrot.cc"

# Inject the required HSTS security header directly into the Nextcloud 443 block
sudo sed -i '/server_name nextcloud.'"${MY_DOMAIN}"';/a \    add_header Strict-Trans
port-Security "max-age=15552000; includeSubDomains; preload;" always;' /etc/nginx/si
tes-available/office.conf

# Test your Nginx configuration syntax
sudo nginx -t

# Reload Nginx to apply the security header immediately
sudo systemctl reload nginx

#####
```

Verify

```
paul@euro-office:~$ podman ps
```

CONTAINER ID	IMAGE	COMMAND	CREA
TED	STATUS	PORTS	NAMES
54d9a1233fc7	docker.io/onlyoffice/documentserver:latest		2 mi
nutes ago	Up 2 minutes	0.0.0.0:8081->80/tcp, 443/tcp	onlyoffice-docs
1bb1d3e6a0a0	docker.io/library/postgres:15-alpine	postgres	2 mi
nutes ago	Up 2 minutes	5432/tcp	nextcloud-db
229512a4eb37	docker.io/library/nextcloud:stable	apache2-foregroun...	About
t a minute ago	Up About a minute	0.0.0.0:8080->80/tcp	nextcloud-app

```
paul@euro-office:~$
```

```
## check Service
journalctl --user -u nextcloud-app.service -n 50 --no-pager
```

Part 5: Initialize the Complete Environment Stack

```
cat << 'EOF' > step5_integration.sh
#!/bin/bash
set -e

echo "=== [Step 5] Initializing Stack & Injecting Proxy Rules ==="

# 1. Establish absolute systemd environment linkages for Debian 13
export XDG_RUNTIME_DIR="/run/user/$(id -u)"
export DBUS_SESSION_BUS_ADDRESS="unix:path=${XDG_RUNTIME_DIR}/bus"

# 2. Part 5: Refresh context and start up the full environment stack cleanly
echo "-> Refreshing systemd user context structures..."
systemctl --user daemon-reload

echo "-> Spawning service definitions..."
systemctl --user start nextcloud-db.service onlyoffice-docs.service nextcloud-app.service

echo "-> Pausing for 30 seconds to let database table migrations settle..."
sleep 30

# 3. Dynamic Loop: Wait until Nextcloud's core interface is completely ready for configurations
echo "-> Checking Nextcloud command-line availability..."
for i in {1..30}; do
    if podman exec -u www-data nextcloud-app php occ status >/dev/null 2>&1; then
        echo "    [+] Nextcloud engine is ready for configurations."
        break
    fi
    if [ "$i" -eq 30 ]; then
        echo "    [!] Error: Nextcloud failed to initialize within expected timeframe s." >&2
        exit 1
    fi
    echo "    [-] Waiting for internal container installation loops (Attempt $i/30)..."
    sleep 5
done

# =====
# Part 6: Reverse Proxy Integration & Domain Whitelisting
# =====

# 1. Allow ONLYOFFICE to query across private container networks
echo "-> Modifying ONLYOFFICE cross-origin private network query rules..."
podman exec -it onlyoffice-docs sed -i 's/"allowPrivateIPAddress": false/"allowPrivateIPAddress": true/g' /etc/onlyoffice/documentserver/default.json
podman exec -it onlyoffice-docs supervisorctl restart all > /dev/null
```

```
# 2. Grant Nextcloud authorization to route local backend loops
echo "-> Allowing Nextcloud to make local internal network calls..."
podman exec -u www-data nextcloud-app php occ config:system:set allow_local_remote_servers --value=true --type=boolean

# 3. Add the Reverse Proxy IP (192.168.0.101) to Nextcloud Trusted Proxies
echo "-> Whitelisting nginx-proxy (192.168.0.101) as a trusted gateway..."
podman exec -u www-data nextcloud-app php occ config:system:set trusted_proxies 0 --value="192.168.0.101"

# 4. Whitelist both local IP paths and the external domain strings
echo "-> Binding trusted local and public domains..."
podman exec -u www-data nextcloud-app php occ config:system:set trusted_domains 1 --value="192.168.0.116"
podman exec -u www-data nextcloud-app php occ config:system:set trusted_domains 2 --value="nextcloud.gardenofrot.cc"
podman exec -u www-data nextcloud-app php occ config:system:set trusted_domains 3 --value="office.gardenofrot.cc"

# 5. Tell Nextcloud to pass web headers correctly through SSL
echo "-> Enforcing SSL proxy headers and routing overwrite flags..."
podman exec -u www-data nextcloud-app php occ config:system:set overwritehost --value="nextcloud.gardenofrot.cc"
podman exec -u www-data nextcloud-app php occ config:system:set overwriteprotocol --value="https"

# 6. Fetch and deploy the official ONLYOFFICE plugin bundle straight into Nextcloud
echo "-> Injecting ONLYOFFICE workspace extension module (Downloading)..."
podman exec -it -u www-data nextcloud-app php occ app:install onlyoffice

echo "=== [Step 5] Script orchestration complete! Environment fully integrated. ==="
EOF

# Make the integration script executable
chmod +x step5_integration.sh

# Run the complete integration script
./step5_integration.sh
```

Verify

```
podman exec -u www-data nextcloud-app php occ config:list system
```

Part 6: Reverse Proxy Integration & Domain Whitelisting

Because traffic passes through the external Nginx reverse proxy, you must instruct both apps to accept requests coming from the domains and proxy IPs.

```
cat << 'EOF' > step6_linkage.sh
#!/bin/bash
set -e

echo "=== [Step 6] Finalising ONLYOFFICE & Nextcloud App Linkage ==="

# 1. Define your domain name environment variable
export MY_DOMAIN="gardenofrot.cc"

# 2. Establish absolute systemd environment linkages for Debian 13
export XDG_RUNTIME_DIR="/run/user/${id -u}"
export DBUS_SESSION_BUS_ADDRESS="unix:path=${XDG_RUNTIME_DIR}/bus"

# 3. Check that the required containers are active before running commands
if ! podman ps | grep -q "nextcloud-app"; then
    echo "    [!] Error: nextcloud-app container is not running. Run step4 and step5
first." >&2
    exit 1
fi

# =====
# SYSTEM CONFIGURATION INJECTION (Direct App Settings)
# =====

echo "-> Setting public ONLYOFFICE Docs Document Server address..."
podman exec -u www-data nextcloud-app php occ config:app:set onlyoffice DocumentServ
erUrl --value="https://office.${MY_DOMAIN}/"

echo "-> Injecting secure JWT authentication secret key..."
# Force overwrites the internal app value to clear browser defaults
podman exec -u www-data nextcloud-app php occ config:app:set onlyoffice DocumentServ
erSecret --value="YourSuperSecretJWTKeyHere"

echo "-> Mapping advanced internal network request routing hooks..."
podman exec -u www-data nextcloud-app php occ config:app:set onlyoffice DocumentServ
erInternalUrl --value="http://onlyoffice-docs/"
podman exec -u www-data nextcloud-app php occ config:app:set onlyoffice StorageUrl
--value="http://nextcloud-app/"

# =====
# CONFIGURATION INTEGRITY VALIDATION
# =====

echo "-> Verifying registered configuration values..."
echo "-----"
podman exec -u www-data nextcloud-app php occ config:app:get onlyoffice DocumentServ
erUrl | sed 's/^/ DocumentServerUrl: /'
podman exec -u www-data nextcloud-app php occ config:app:get onlyoffice DocumentServ
erInternalUrl | sed 's/^/ DocumentServerInternalUrl: /'
```

```
podman exec -u www-data nextcloud-app php occ config:app:get onlyoffice StorageUrl |
sed 's/^/  StorageUrl: /'
echo "-----"

echo "=== [Step 6] Script orchestration complete! Sovereign Office suite is fully li
ve. ==="
EOF

# Ensure script execution rights are preserved
chmod +x step6_linkage.sh

# Run the updated, variable-driven linkage script
./step6_linkage.sh
```

Part 7: Finalize Integration in Web Browser

```
# get ADMIN password
podman exec -it nextcloud-app env | grep NEXTCLOUD_ADMIN_PASSWORD
```

open <https://nextcloud.gardenofrot.cc/apps/dashboard/> (<https://nextcloud.gardenofrot.cc/apps/dashboard/>)

Admin panel is not need as it all on one server

1. Open the browser and go to the public address: <https://yourdomain.com> (<https://yourdomain.com>)
2. Authenticate using the deployment credentials (admin / YourSecureAdminPassword).
3. Click the profile icon (top right corner) → Administration settings.
4. Select ONLYOFFICE on the bottom left navigation tree menu.
5. Expand the Advanced server settings tab to bypass proxy loop limits:
 - ONLYOFFICE Docs address: <https://yourdomain.com> (<https://yourdomain.com>)
 - Secret key: YourSuperSecretJWTKeyHere
 - ONLYOFFICE Docs address for internal requests from the server: <http://onlyoffice-docs/> (<http://onlyoffice-docs/>)

- Nextcloud address for internal requests from the document server: <http://nextcloud-app/> (<http://nextcloud-app/>)

6. Press Save. the collaborative workspace environment is ready.

Part 3: Nextcloud Integration & Post-Deployment Tuning

Once the Podman Quadlets are running and your Nginx configuration blocks are enabled, you must link the applications. Follow these steps to configure the integration, resolve Mixed Active Content errors, and prevent app framework conflicts.

1. Whitelist Local Loopback Container Networks

By default, Nextcloud blocks outgoing HTTP requests destined for local subnets or private address spaces. Run this command inside your application terminal to allow Nextcloud to talk directly to your ONLYOFFICE container via the native Podman network mesh:

```
podman exec --user www-data nextcloud-app php occ config:system:set allow_local_remote_servers --value=true --type=boolean
```

2. Enable the Integration Connector App

1. Log into your Nextcloud instance web UI as an **Administrator**.
2. Click your profile icon in the top-right corner and select **Apps**.
3. Locate **ONLYOFFICE** under disabled apps (or search for it using the magnifying glass tool) and select **Download and enable**.

3. Split-Horizon URI Network Configuration

Navigate to **Administration settings** -> **ONLYOFFICE** via the left-hand sidebar menu. To bypass browser mixed content blocks while routing heavy file traffic locally inside the network, populate the advanced fields exactly as shown below:

Server Settings Block

- **ONLYOFFICE Docs address:** `https://gardenofrot.cc` (Note: This is the external HTTPS URI accessed by your web browser)
- **Secret key:** `YourSuperSecretJWTKeyHere` (Note: Must exactly match the `JWT_SECRET` environment flag declared in your Quadlet container file)

Advanced Server Settings Dropdown

- **ONLYOFFICE Docs address for internal requests from the server:** `http://onlyoffice-docs/` (Note: Routes deep-backend API callbacks across the internal network mesh using unencrypted HTTP)
- **Server address for internal requests from ONLYOFFICE Docs:** `http://nextcloud-app/` (Note: Directs ONLYOFFICE to fetch the target binary assets directly from Nextcloud's local port)

Click the blue **Save** button. The document parameters will populate downward on successful validation. Ensure the check box next to `docx` (along with `xlsx` or `pptx`) is enabled.

?? Deactivate Overlapping Layout Frameworks

To ensure Nextcloud correctly routes web editing tasks away from alternative engines like Collabora/Nextcloud Office, deactivate conflicting document apps:

1. Navigate to **Administration settings** -> **Nextcloud Office**.
2. Scroll to the section titled **The default application for opening the format**.
3. **Uncheck** the parameter box for `docx` to hand layout rendering entirely to ONLYOFFICE. Project Screenshot

Part 8: Master script and readme.

```
cat << 'EOF' > master_deploy.sh
#!/bin/bash
set -e

echo "=====
echo "🚀 STARTING MASTER DEPLOYMENT: SOVEREIGN DIGITAL WORKSPACE"
echo "=====

# 1. Establish absolute systemd environment linkages for Debian 13
export XDG_RUNTIME_DIR="/run/user/$(id -u)"
export DBUS_SESSION_BUS_ADDRESS="unix:path=${XDG_RUNTIME_DIR}/bus"

# Cache sudo credentials upfront so the script runs unattended
sudo -v

# 2. Check for required local static configuration files
FILES=("nextcloud-db.container" "onlyoffice-docs.container" "nextcloud-app.container")
for file in "${FILES[@]"; do
    if [ ! -f "$file" ]; then
        echo "    [!] Error: Critical file '$file' missing from working directory." >
&2
        echo "        Please execute ./step4_quadlets.sh once to populate static asse
ts." >&2
        exit 1
    fi
done

# 3. Sequential Orchestration Pipeline
echo ""
echo "👉 Running Step 3: Initializing Storage Infrastructure..."
./step3_storage.sh

echo ""
echo "👉 Running Step 4: Compiling & Syncing Quadlet Service Matrix..."
./step4_quadlets.sh

echo ""
echo "👉 Running Step 5: Provisioning Domain Whitelists & Extension Engine..."
./step5_integration.sh

echo ""
echo "👉 Running Step 6: Finalizing Cryptographic Application Handshakes..."
./step6_linkage.sh

echo ""
echo "=====
echo "🎉 SUCCESS: Environment is fully operational and tracked in version control!"
echo "=====
```

```
podman ps
EOF

# Make the master script executable
chmod +x master_deploy.sh
```

master readme.md

```
cat << 'EOF' > README.md
# Sovereign Collaborative Workspace Setup Guide
### Nextcloud Hub & ONLYOFFICE Docs with Rootless Podman and Systemd Quadlets

This repository provides an production-ready blueprint to deploy an automated collaborative office suite. It utilizes rootless Podman, Debian-native Quadlets for systemd process management, and an external Nginx load-balancer for SSL termination and secure WebSocket routing.

---

## 🚧 Infrastructure Architecture
```

Maintenance & Lifecycle Commands

- Stop Ecosystem: `systemctl --user stop nextcloud-app onlyoffice-docs nextcloud-db`
 - Start Ecosystem: `systemctl --user start nextcloud-db onlyoffice-docs nextcloud-app`
 - Restart Ecosystem: `systemctl --user restart nextcloud-db onlyoffice-docs nextcloud-app`
 - Review Live Event Logs: `journalctl --user -u nextcloud-app.service -f`
-

Would you like help generating Let's Encrypt SSL certificates via certbot on the Nginx proxy server, or would you like to verify the internal Podman logs if a container fails to start?